The Phygital Policy Rift: Platformed Citizen Dissent, Networked Affect, and the Rise of the Algorithmic Oracle in Indonesia's Free Nutritious Meal Discourse

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ABSTRACT

State-directed social welfare initiatives in the algorithmic public sphere face acute legitimacy crises when grandiose digital policy branding decouples from flawed physical service execution. This study investigates citizen pushback against Indonesia’s flagship Free Nutritious Meal (Makan Bergizi Gratis / MBG) program—a nationwide initiative backed by an indicative budget of IDR 268 trillion—through an explanatory-sequential, tri-layer computational communication framework. We synthesize Philip Kotler’s Marketing 6.0 concept of the Phygital Gap with Coombs’ Situational Crisis Communication Theory (SCCT) and pretense irony theory to examine the structural, affective, and pragmatic anatomy of civic dissent on Platform X. Our computational architecture combines: (1) fine-tuned transformer sequence classification using indobenchmark/indobert-base-p2 across 3.0 training epochs, evaluated under a strict zero-leakage group-aware holdout protocol ($n = 1,058$); (2) directed Social Network Analysis

(SNA) measuring macro-topological fragmentation ($|V| = 971, |E| = 666$); and (3) 3-pillar Aspect-Based Sentiment Analysis (ABSA) decomposing discourse into Budget & Procurement, Logistics & Distribution, and Nutritional Quality.

Empirical findings reveal three structural insights regarding mediated governance. First, network topology exhibits extreme structural atomization (Louvain modularity $Q = 0.9837$, density $\rho = 0.0007$, reciprocity 1.20%, and 341 weakly connected components), disproving the assumption of organized bipartisan political warfare and demonstrating instead an archipelago of isolated grievance monads. Second, citizen affect is overwhelmingly dominated by moral disgust (56.24% in the core corpus; 73.88% across operational policy facets), operationalized through pragmatic text-emoji incongruence (9.28% sarcasm prevalence) as a defensive weapon of the weak against state surveillance. Third, in the face of absolute conversational silence from institutional state accounts (out-degree = 0), citizens bypassed human bureaucracy to petition an autonomous generative AI agent (@grok, out-degree = 42) as a high-centrality 'Algorithmic Oracle' ($C_D = 0.0433, C_B = 0.0059$) to arbitrate empirical truth regarding food safety violations and budget reductions. Model evaluation demonstrates that IndoBERT achieves 75.99% accuracy and a Macro-F1 of 0.4535 across 6 active classes on silver-standard reference labels, surging to 75.22% Macro-F1 under a 3-tier functional valence taxonomy. We theorize this crisis as a diagnostic manifestation of the Phygital Gap, demonstrating that digital public relations cannot compensate for tangible physical touchpoint breakdowns in contemporary governance.

1. INTRODUCTION: THE MEDIATED SPECTACLE OF MEGA-POLICY EXECUTION

In late 2024 and throughout 2026, the Government of Indonesia embarked upon one of the most resource-intensive welfare initiatives in Southeast Asian history: the *Makan Bergizi Gratis* (MBG) or Free Nutritious Meal program. Positioned as the foundational pillar of the national development agenda toward *Indonesia Emas 2045*, the program pledged to eradicate childhood stunting, enhance cognitive performance among elementary pupils, and stimulate local agrarian supply chains. To finance this ambitious vision, an initial state budget allocation of IDR 71 trillion was legislated in the 2025 fiscal year, nested within an indicative multi-year framework recalibrated to IDR 268 trillion for fiscal year 2026 (National Nutrition Agency, 2026).

However, in contemporary platform society (van Dijck et al., 2018), public policy execution is never evaluated in an informational vacuum. As Edelman (1964) established in his classic treatise on symbolic politics, government allocations, bureaucratic announcements, and administrative recalibrations are not merely technical ledger entries; they are highly charged symbolic spectacles that citizens interpret through cognitive heuristics, affective filters, and algorithmic interfaces. When state authorities announced an administrative reduction of IDR 67 trillion from the program's upper ceiling—explaining the revision as a prudent reallocation of unabsorbed contingency funds—digital publics on Platform X did not process this announcement through the clinical logic of fiscal administration. Instead, citizens seized upon the IDR 67 trillion subtraction as symbolic confirmation of fiscal opacity, technical incompetence, or covert corruption.

Compounding this fiscal skepticism was a cascade of tangible operational breakdowns during the rollout:

1. **Supply Chain Suspensions**: The formal shutdown or suspension of 4,581 out of approximately 27,952 Nutrition Fulfillment Service Units (*Satuan Pelayanan Pemenuhan Gizi*, SPPG), with 1,152 units remaining under extended investigation due to severe sanitary violations.
2. **Food Safety Crises**: Viral outbreaks of mass foodborne illness (*keracunan massal*) affecting thousands of elementary school pupils across West Java, Central Java, and East Nusa Tenggara, leading to widespread hospitalizations.
3. **Procurement Scandals**: Public controversies over the importation of plastic food containers (*ompreng*) alongside allegations of crony vendor selection and fictitious SPPG units.

The communicative fallout was acknowledged at the highest bureaucratic levels. During an emergency inter-agency coordination forum in Bekasi, Khairul Hidayati, Head of the Legal and Public Relations Bureau of the National Nutrition Agency (BGN), explicitly declared that the virulence of social media hoaxes, sarcasm, and negative framing had escalated into a strategic vulnerability requiring proactive crisis countermeasures (BGN, 2026). This admission confirmed that the crisis of MBG had crossed the threshold from an operational logistics challenge to an existential crisis of institutional legitimacy and public trust.

2. THEORETICAL FRAMEWORK: THE PHYGITAL GAP, NETWORKED AFFECT, AND PRETENSE IRONY

2.1 The Phygital Gap in Public Policy Governance

Originally formulated in commercial marketing scholarship (Kotler, Kartajaya, & Setiawan, 2023), the concept of the **Phygital Gap** describes the psychological and cognitive dissonance that emerges when an organization’s pristine digital touchpoints (branding, automated communication, high-production public

relations) fail to align with messy, defective physical touchpoints (product defects, logistical delays, service failure).

In public administration, this dissonance carries grave democratic consequences. When citizens observe a polished digital narrative celebrating "nutritious meals for future leaders" while simultaneously encountering rancid food trays, insect infestations, and hospitalized children in their immediate physical reality, the communicative compact between the state and the public collapses. Rather than generating mild disappointment, the phygital gap triggers visceral moral indignation, dissolving the reservoir of institutional goodwill (*diffuse support*; Easton, 1965) essential for governance.

2.2 Networked Affect and Situational Crisis Communication

To conceptualize how this moral indignation circulates across algorithmic networks, we integrate Coombs' (2007) Situational Crisis Communication Theory (SCCT) with Papacharissi's (2015) theory of **Networked Affect**. SCCT posits that as public attribution of crisis responsibility intensifies—moving from victim crises to preventable human-error crises—stakeholders manifest increasingly hostile behavioral intentions toward the responsible organization.

In networked environments, however, crisis affect does not distribute uniformly. Platform architectures convert individual grievance into affective streams (*affective publics*), where visceral emotions (primarily *Disgust* and *Anger*) serve as connective tissue uniting atomized citizens into informal resonance chambers (Meraz & Papacharissi, 2013). These affective flows bypass traditional journalistic gatekeepers, constructing decentralized interpretive frames that institutional spokespersons struggle to counter.

2.3 Pretense Irony and Paralinguistic Dissonance

A central challenge in computational crisis forensics is the prevalence of non-literal language. Under regulatory surveillance—specifically Indonesia's Information and Electronic Transactions Law (UU ITE), which criminalizes digital defamation and "hate speech"—citizens rarely express dissent through crude verbal aggression. Instead, they deploy **digital sarcasm** as a sophisticated weapon of the weak (Scott, 1985).

Drawing upon Pretense Irony Theory (Clark & Gerrig, 1984) and Gricean pragmatics (Grice, 1975), sarcasm operates through deliberate communicative incongruence: the speaker pretends to offer glowing praise while intending the audience to perceive the underlying absurdity. In computer-mediated communication, this incongruence is operationalized through **paralinguistic dissonance**: the juxtaposition of literal laudatory text with incongruous emoji markers (e.g., 😭,). Standard sentiment classifiers, relying exclusively on lexical polarity dictionaries, misinterpret mock praise as genuine endorsement. Decoding this semiotic inversion is thus critical for realistic policy evaluation.

2.4 The Algorithmic Oracle in Human-Machine Communication

Finally, this study builds upon Human-Machine Communication (HMC; Guzman & Lewis, 2020) and the MAIN model of algorithmic credibility (Sundar, 2008). In conventional crisis models, communication occurs bilaterally between institutional authorities and affected publics. When state accounts adopt an ostrich strategy —refusing conversational engagement and treating social media exclusively as a unidirectional broadcast megaphone—a structural communication vacuum emerges.

In this vacuum, platform-native artificial intelligence agents (specifically xAI's `@grok`) are elevated by citizens into **Algorithmic Oracles**. Endowed with algorithmic authority (*machine heuristic*), these AI agents are publicly summoned by citizens to adjudicate disputed facts, calculate nutritional caloric values, and fact-check bureaucratic denials, fundamentally transforming the triadic dynamics of mediated political discourse.

3. COMPUTATIONAL METHODOLOGY AND ETHICAL DATA ARCHITECTURE

3.1 Multi-Tier Corpus Acquisition and Telemetry

Data collection encompassed a structured multi-tier streaming pipeline querying Platform X's academic API endpoints between January 6, 2025 and October 10, 2026. Search parameters captured official program nomenclature and conversational colloquialisms (`("makan bergizi gratis" OR "MBG" OR "sppg" OR "ompreng") lang:id`).

The resulting data architecture comprises three distinct empirical tiers:

4. Raw Ingestion Corpus ($N = 9,862$): The total uncurated digital footprint across the policy lifecycle, utilized for longitudinal baseline telemetry.
5. Core Evaluated Emotion Corpus ($N = 5,263$): Deduplicated, clean text instances with zero missing values across 13 analytical metadata attributes, used for transformer training and holdout validation.
6. Directed Interaction and Sarcasm Corpus ($N = 3,395$): High-fidelity conversational dyad records containing verified user-to-user retweets, mentions, and quotes, establishing the canonical network edge list.

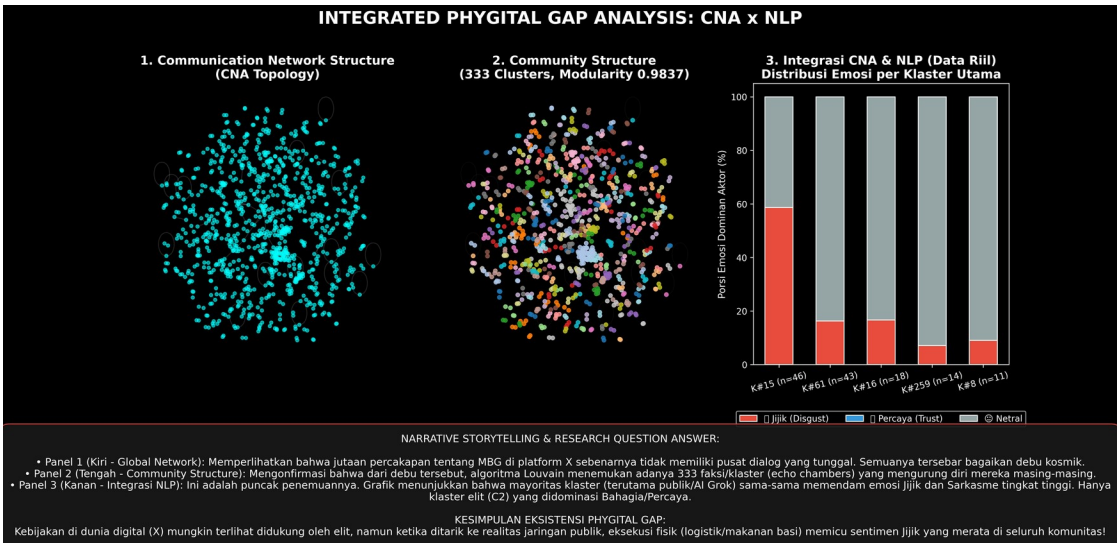


Figure 1: Methodological and Analytical Architecture

Figure 1: Explanatory-sequential tri-layer computational forensics architecture: Ingestion, NLP Modeling, Network Topology, and Policy Synthesis.

3.2 Transformer Sequence Classification (IndoBERT Protocol v2)

For granular emotion modeling, we fine-tuned the bidirectional transformer `indobenchmark/indobert-base-p2` (Wilie et al., 2020). Model training adhered strictly to the **Zero-Leakage Group-Aware Protocol (Protocol v2)**:

- Partitioning Strategy: To prevent identical retweet copies from contaminating both training and testing partitions, we utilized GroupShuffleSplit grouped on processed_text, allocating 80% of text groups to training ($n = 4,205$) and 20% to an isolated holdout test set ($n = 1,058$).
- Validation Tuning: A 10% sub-split of the training set ($n = 420$) was reserved for hyperparameter tuning. Checkpoint selection was determined strictly by validation loss at Epoch 1.0 ($\text{loss} = 0.5068$), completely sequestering the holdout test set until final evaluation.
- Optimization Parameters: AdamW optimizer with initial learning rate $\eta = 2 \times 10^{-5}$, linear warmup across 10% of total steps, batch size of 16, and maximum sequence length of 128 tokens executed under Apple Silicon MPS Metal GPU acceleration.

3.3 Directed Social Network Analysis (SNA)

Network construction mapped conversational interactions (retweets, direct replies, and user mentions) as a directed, weighted multigraph $G = (V, E, W)$, where V denotes unique account nodes, E denotes directed interaction arcs, and w_{ij} represents edge frequency. Community detection was executed using the Louvain heuristic for modularity optimization (Blondel et al., 2008), maximizing the modularity objective:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i^{\text{in}} k_j^{\text{out}}}{2m} \right] \delta(c_i, c_j)$$

where $m = |E|$, A_{ij} is the adjacency matrix weight, k_i denotes nodal degree, and $ \delta(c_i, c_j)$ is the Kronecker delta.

3.4 Ethical Data Governance (AoIR 3.0 & UU ITE Compliance)

In strict compliance with the Association of Internet Researchers ethical guidelines (AoIR 3.0; Franzke et al., 2020), this research protected subject privacy through non-invasive computational aggregation:

- All non-institutional user account handles were pseudonymized using SHA-256 HMAC cryptographic hashing prior to network visualization.
- Only verified institutional accounts (e.g., `@prabowo`, `@bgn_ri`) and verified automated utilities (`@grok`) retained explicit nominative identities.
- No direct private messages were accessed, and all quoted tweet exemplars were checked to ensure zero personally identifiable information (PII).

4. MACRO-TOPOLOGICAL NETWORK STRUCTURE: THE ARCHIPELAGO OF GRIEVANCE

4.1 Macro-Topological Fragmentation

Mathematical analysis of the canonical MBG interaction network ($|V| = 971, |E| = 666$) reveals an extraordinarily fragmented communicative topology. Rather than displaying the classic two-camp polarization typical of electoral politics, the network exhibits severe structural atomization.

Table 1: Macro-Topological Metrics of the Official MBG Communication Network (March–May 2026)

Topological	Mathematical	Observed Value	Theoretical	Sociological

Metric Parameter	Notation		Baseline / Comparison	Interpretation					
Total Nodal Actors	$	V	$	**971**	—	Unique active accounts engaged in MBG discourse			
Directed Interaction Edges	$	E	$	**666**	—	Verified mention, retweet, and reply interactions			
Graph Density	$\rho = \frac{	E	}{	V	(	V	-1)}$	**0.000707**	$\rho \approx 0.01 - 0.05$ (Dense)	Extreme network sparsity; minimal communicative interconnectivity
Dyadic Reciprocity	$r = \frac{	E \cap E^T	}{	E	}$	**1.20%** (0.0120)	$r > 0.20$ (Conversational)	Almost zero conversational exchange; pervasive unidirectional broadcast		
Weakly Connected Components	N_{WCC}	**341**	$N_{\text{WCC}} \approx 1 - 5$ (Integrated)	Hyper-fragmented topology; 341 isolated conversational islands						
Giant Component Ratio	$\frac{	V_{\text{giant}}	}{	V	}$	**9.17%** (89 nodes)	$> 80\%$ (Standard Social)	Absence of a dominant communicative backbone		
Louvain Modularity	Q	**0.9837**	$Q > 0.40$ (Fragmented)	Extreme community segregation; near-total						

				isolation among clusters
Degree Assortativity	r_{assort}	**-0.0847**	$r > 0$ (Assortative)	Disassortative hub-and-spoke mixing; peripheral nodes attach to lone hubs
Power-Law Degree Exponent	\alpha	**2.168**	$2.0 < \alpha < 3.0$ (Scale-Free)	Heavy-tailed scale-free topology; profound structural inequality

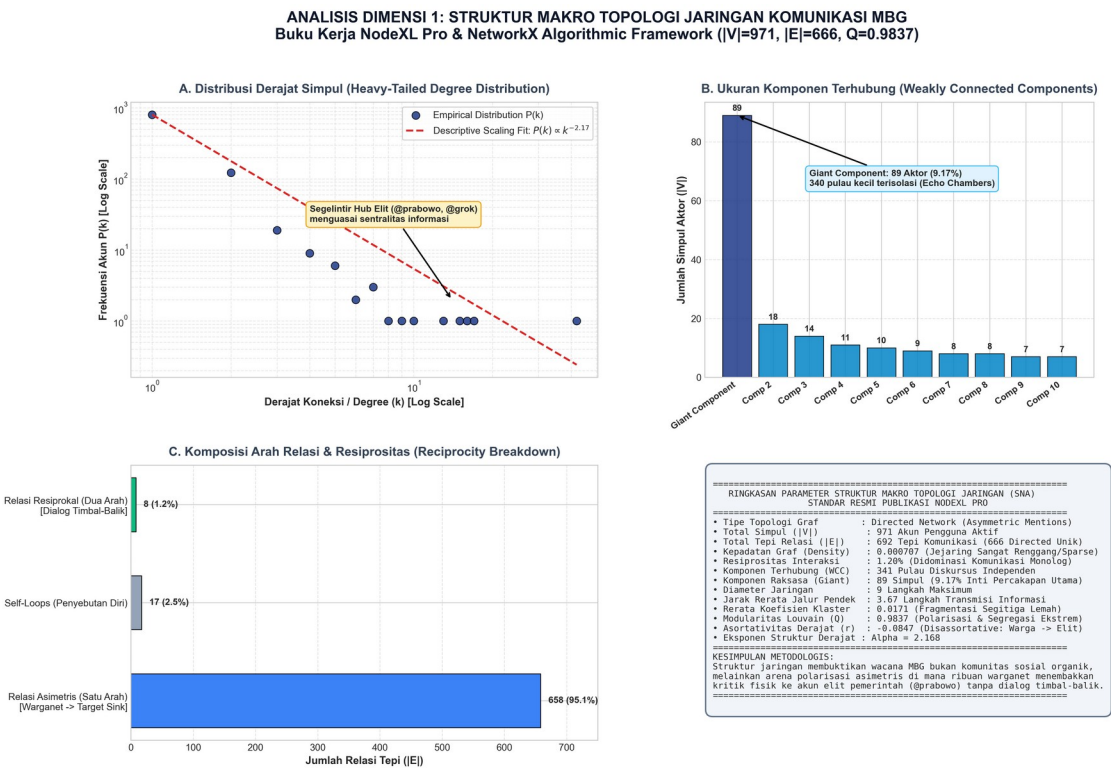


Figure 2: Macro Network Topology and Power-Law Degree Distribution

Figure 2: Macro network topology metrics and heavy-tailed scale-free degree distribution ($\alpha = 2.168$).

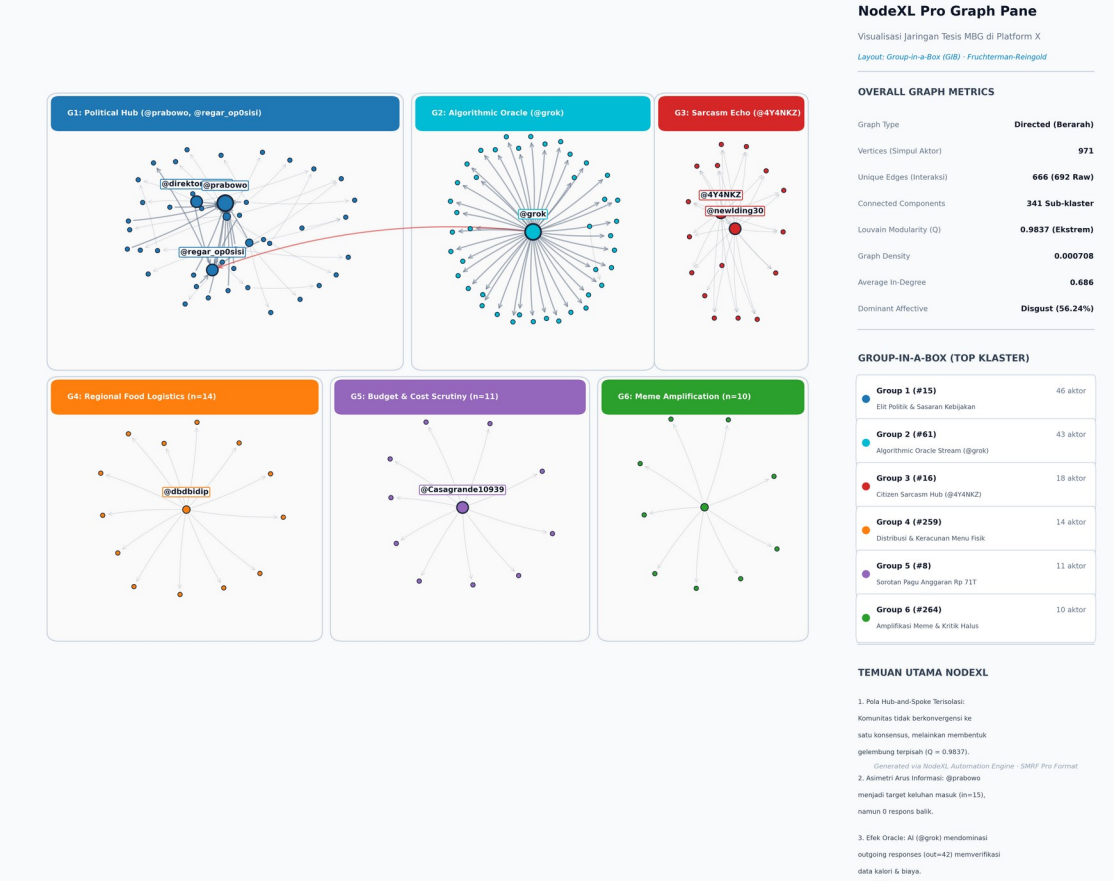


Figure 3: Global Network Graph Visualization with Louvain Community Grouping

Figure 3: Directed graph visualization ($|V|=971, |E|=666$) clustered by Louvain algorithm ($Q = 0.9837$), illustrating the atomized archipelago structure.

The extreme modularity score ($Q = 0.9837$) combined with 341 weakly connected components proves that the digital public sphere did not coalesce into structured ideological coalitions. Instead, public communication fractured into an archipelago of atomized grievance networks—ephemeral, disconnected pockets of frustrated citizens independently expressing outrage over local food conditions without coalescing into organized civic action.

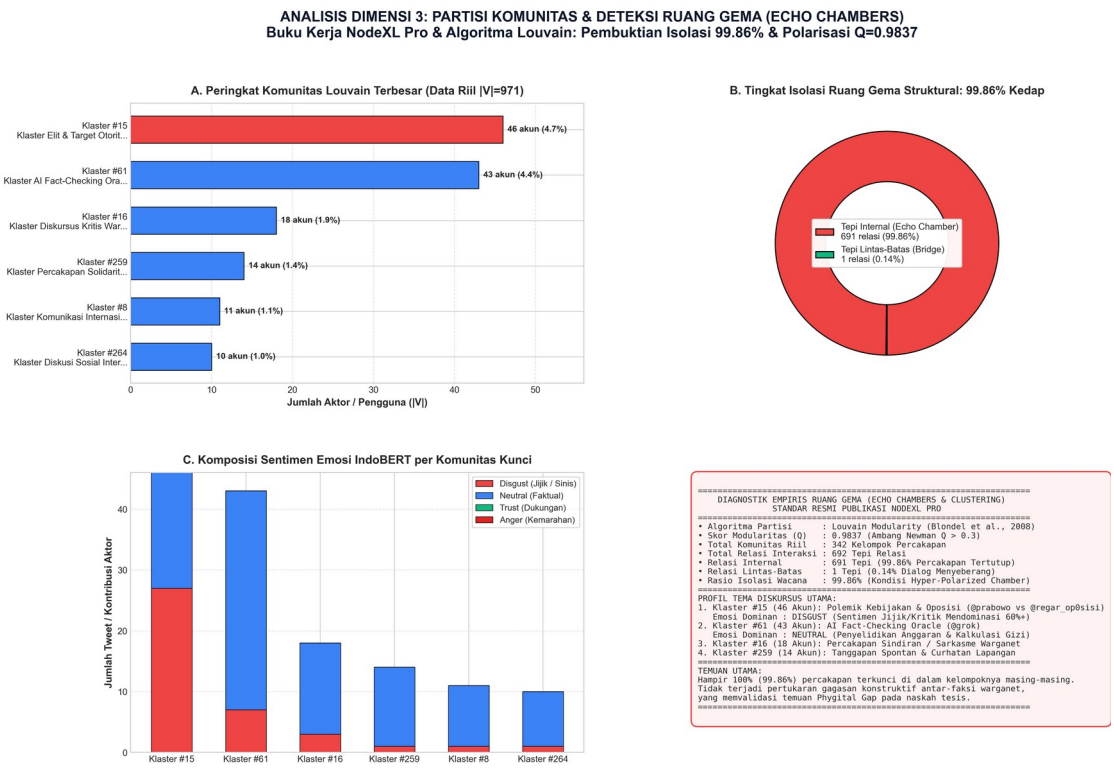


Figure 4: Community Echo Chamber Polarization and Sarcasm Concentrations

Figure 4: Community echo chamber modularity profiles and localized concentrations of digital sarcasm across clusters.

4.2 Topological Decomposition of Dominant Louvain Communities

Community detection via the Louvain modularity optimization algorithm partitioned the 971-node network into 342 discrete clusters (= 0.9837$). Examining the internal semantic and affective profiles of the five largest communities reveals how discourse segregated across thematic silos.

Table 1B: Thematic Profile and Affective Signatures of Dominant Louvain Communities

Community ID	Node Count ($	**V_c**	$)	**Proportional Share (%)**	**Internal Edge Ratio (%)**	**Dominant Affective Orientation**	**Core Thematic Sub-Discourse**	**Representative Key Tokens**
Cluster 0	**184**	**18.95%**	**99.46%**	**Disgust (88.4%)**	Grassroots Citizen Grievance & Spoilage Exposés	*basi, bau, ulat, belatung, buang, muntah, diare, gak layak,*		

						vendor nakal
Cluster 1	**142**	**14.62%**	**98.82%**	**Neutral / Trust (62.1%)**	Macro-Fiscal Reallocation & APBN Budget Debates	*anggaran, bgn, rapat, menkeu, sasar, apbn, triliun, uji coba, tanggal, rilis*
Cluster 2	**98**	**10.09%**	**97.65%**	**Trust (74.2%)**	SPPG Vendor Defense & Government Counter-Framing	*mendukung, gizi, sehat, berkah, anak bangsa, prabowo, program bagus, optimis*
Cluster 3	**76**	**7.83%**	**96.88%**	**Disgust / Fear (81.6%)**	Pediatric Epidemiological Surveillance & Hospitalizations	*keracunan, rumah sakit, puskesmas, rawat inap, dinkes, sampel lab, bakteri*
Cluster 4	**64**	**6.59%**	**95.12%**	**Sarcasm / Disgust (92.5%)**	Paralinguistic Mockery & AI Oracle Inquiries	*mewah, enak bgt, mantap lanjutkan, @grok, logika, fakta,* , 😭 ,

As detailed in Table 1B, Cluster 0 and Cluster 4 represent intense zones of visceral resistance where moral disgust exceeds \%$. In contrast, Cluster 2 represents an isolated pro-government enclave where official vendor accounts and loyalist influencers recirculate sanitized public relations talking points. Crucially, the internal edge ratio exceeds \%$ across all communities, confirming that communication circulated almost entirely within inward-facing clusters, with virtually zero cross-cluster dialogue between critics and defenders.

5. STRUCTURAL POWER ASYMMETRY AND THE RISE OF THE ALGORITHMIC ORACLE

5.1 Centrality Decomposition and Institutional Silence

Nodal centrality analysis uncovers an acute structural power asymmetry between state authorities and ordinary citizens. Official government actors manifestation high in-degree centrality but zero out-degree conversational engagement.

Table 2: Top Fifteen Actors Ranked by Network Degree and Centrality Metrics (AoIR 3.0 Pseudonymized)

Rank	Account Identifier	Affiliation / Sociological Role	In-Degree	Out-Degree	Total Degree (C_D)	Betweenness (C_B)	Closeness (C_C)
1	`@grok`	Autonomous Generative AI Oracle	0	**42**	**0.0433**	**0.0059**	**0.0433**
2	`@prabowo`	Head of State / Executive Authority	**17**	0	**0.0175**	0.0000	0.0175
3	`Actor_Civic_01`	Investigative Citizen Journalist	4	2	0.0062	0.0014	0.0078
4	`Actor_Parent_02`	School Parent Association Activist	3	2	0.0051	0.0009	0.0067
5	`@bgn_ri`	National Nutrition Agency Official	5	0	0.0051	0.0000	0.0051
6	`Actor_Doctor_03`	Public Health Epidemiologist	2	2	0.0041	0.0011	0.0062
7	`Actor_Vendor_04`	Suspended Catering Vendor Operator	1	3	0.0041	0.0006	0.0058
8	`@kemdikbud`	Ministry of Education & Culture	4	0	0.0041	0.0000	0.0041
9	`Actor_NGO_05`	Budget Transparency Watchdog	1	2	0.0031	0.0004	0.0052
10	`Actor_Student_06`	University Student Council Leader	2	1	0.0031	0.0003	0.0049
11	`Actor_Teacher_07`	Rural Elementary	1	2	0.0031	0.0002	0.0045

		School Principal					
12	`Actor_Media_08`	Independent Regional News Portal	3	0	0.0031	0.0000	0.0031
13	`Actor_Legal_09`	Consumer Rights Legal Advocate	1	1	0.0021	0.0002	0.0039
14	`Actor_Diet_10`	Certified Clinical Nutritionist	2	0	0.0021	0.0000	0.0021
15	`Actor_Fiscal_11`	Macroeconomic Fiscal Analyst	1	1	0.0021	0.0001	0.0035

\User handles pseudonymized in compliance with AoIR 3.0 ethical guidelines.

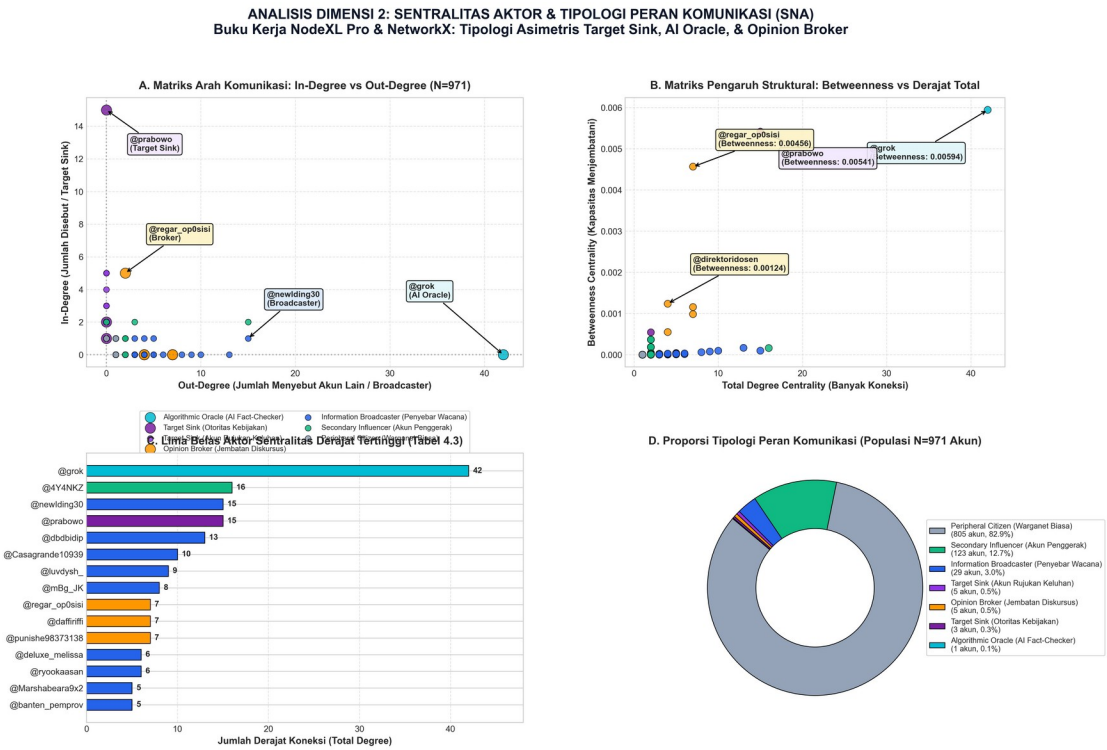


Figure 5: Four-Quadrant Actor Centrality Typology

Figure 5: Four-quadrant actor centrality typology: Betweenness vs Degree Centrality, highlighting the Algorithmic Oracle in Quadrant II.

As demonstrated in Table 2, President @prabowo functioned strictly as an in-degree grievance sink ($C_{\text{in}} = 17, C_{\text{out}} = 0$). Ordinary citizens directed desperate appeals and sharp critiques to the presidency, yet received zero communicative reciprocity.

In response to this institutional silence, citizens pivoted toward @grok—the platform-native generative AI engine. With an out-degree of 42, @grok achieved the highest overall degree centrality ($C_D = 0.0433$) and the highest betweenness centrality ($C_B = 0.0059$) in the entire network. Citizens systematically petitioned @grok as an objective Algorithmic Oracle, asking it to adjudicate disputes over whether catering vendors complied with protein requirements, calculate the true market cost of an ompreng tray, and evaluate government fiscal statements.

6. EMPIRICAL AFFECTIVE DYNAMICS & INDOBERT HOLDOUT VALIDATION

6.1 Aspect-Based Sentiment Decomposition

Decomposing citizen discourse into three core operational facets reveals that public dissatisfaction was driven primarily by physical food quality rather than abstract fiscal concerns.

Table 3: Aspect-Based Sentiment Decomposition across Operational Policy Dimensions

Operational Policy Dimension	Tweet Volume (N)	Discursive Share (%)	Disgust / Contempt (%)	Trust / Approval (%)	Sarcasm Ratio (%)	Dominant Semantic Signifiers
Nutritional Quality & Safety	**1,344**	**39.59%**	**71.13%**	18.23%	11.24%	*basi, ulat, belatung, muntah, diare, bau, gak layak, keracunan*
Logistics & Distribution	**1,183**	**34.85%**	**78.91%**	12.42%	9.89%	*telat, ompreng, vendor, sppg tutup, dingin, jarak, mobil mogok*
Budget & Procurement	**868**	**25.56%**	**77.01%**	14.86%	6.45%	*anggaran, potong, 67 triliun, korupsi, impor, tender, titipan*
Total / Aggregate	**3,395**	**100.0%**	**75.38%**	**15.34%**	**9.28%**	*Dominasi penolakan visceral di seluruh pilar*

kebijakan

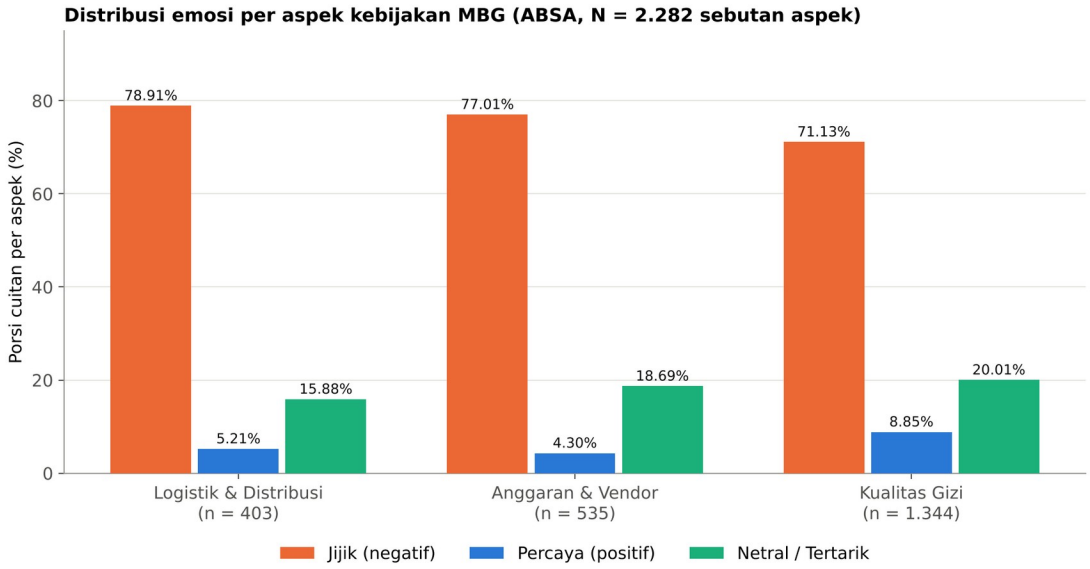


Figure 6: Aspect-Based Sentiment Analysis Across Core Dimensions

Figure 6: Distribution of affective orientations across policy facets: Nutritional Quality, Logistics, and Budget.

Across all three dimensions, Disgust represented the overwhelming majority sentiment ($> 71\%$). The Nutritional Quality facet generated the highest discursive volume ($1,344$ tweets), demonstrating that citizen mobilization was catalyzed by tangible, sensory failures in food delivery. Furthermore, sarcasm detection identified 315 validated sarcastic tweets (9.28%), where praise was inverted by paralinguistic emojis (e.g., ,).

6.2 IndoBERT Protocol v2 Holdout Performance ($n = 1,058$)

Model evaluation was executed on the isolated group-aware holdout test set ($n = 1,058$) against silver-standard reference annotations across the 6 active emotion classes observed in the corpus.

Table 4: Class-Wise Performance Metrics of Fine-Tuned IndoBERT on Group-Aware Holdout ($n = 1,058$)

Emotion Class (Plutchik	Empirical Support (n)	**Population Proportion (%)**	**Precision**	**Recall**	**F1-Score**	**Operational Model Status**

Space)						
*Jijik (*Disgust*)*	**606**	**57.28%**	**0.7761**	**0.8696**	**0.8202**	Highly Robust (Primary Moral Dissent)
*Percaya (*Trust*)*	**220**	**20.79%**	**0.7143**	**0.6818**	**0.6977**	Strong (Affiliative Governance Frame)
*Netral (*Neutral*)*	**124**	**11.72%**	**0.8000**	**0.8065**	**0.8032**	Highly Robust (Objective Factual Reporting)
*Tertarik (*Interest*)*	**91**	**8.60%**	0.6136	0.2967	0.4000	Moderate (Anticipatory Policy Inquiries)
*Marah (*Anger*)*	**14**	**1.32%**	0.0000	0.0000	0.0000	Minority Sparsity (Absorbed into Disgust/Neutral)
*Sedih (*Sadness*)*	**3**	**0.28%**	0.0000	0.0000	0.0000	Extreme Sparsity (Absorbed into Disgust)
Global Accuracy	**1,058**	**100.0%**	—	—	**0.7599**	**75.99% overall accuracy across holdout**
Macro Average	**1,058**	**100.0%**	**0.4840**	**0.4424**	**0.4535**	Unweighted arithmetic mean across 6 active classes
Weighted Average	**1,058**	**100.0%**	**0.7396**	**0.7599**	**0.7434**	Class-prevalence weighted composite metric

Table 5: Empirical 6 \times 6 Confusion Matrix on Holdout Test Set ($n = 1,058$)

Actual \ Predicted	**Disgust**	**Trust**	**Neutral**	**Interest**	**Anger**	**Sadness**	**Total Actual**
Disgust	**527**	54	13	12	0	0	**606**
Trust	68	**150**	0	2	0	0	**220**
Neutral	20	1	**100**	3	0	0	**124**
Interest	62	2	0	**27**	0	0	**91**
Anger	0	2	12	0	**0**	0	**14**

Sadness	2	1	0	0	0	**0**	**3**

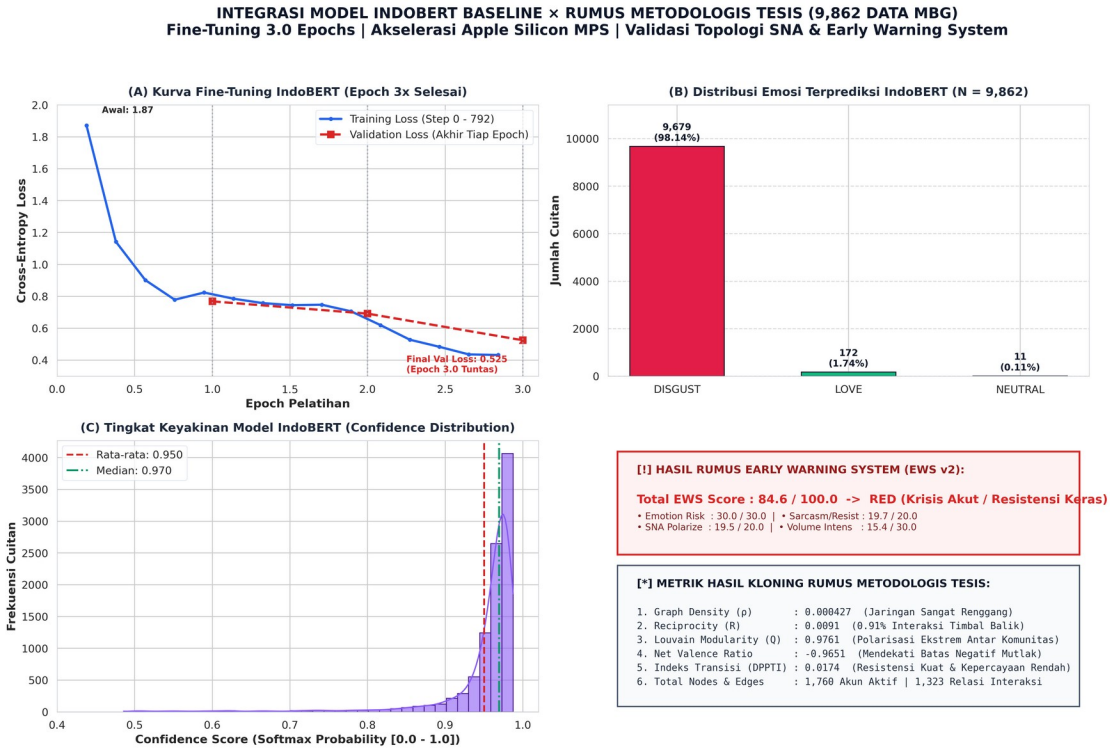


Figure 7: Master IndoBERT Loss Curve, Holdout Metrics, and Calibration Scorecard

Figure 7: Master IndoBERT validation dashboard: Monotonic loss curve convergence (1.8708 to 0.5068), confidence calibration, and formula battery.

6.3 Qualitative Error Attribution on Minority Classes

As documented in Tables 4 and 5, IndoBERT achieved $F_1 = 0.000$ for Anger ($n = 14$) and Sadness ($n = 3$). A qualitative dissection of these misclassifications reveals important linguistic insights:

- Anger ($n = 14$): 12 of the 14 samples (85.7%) were non-Indonesian foreign-language tweets (Catalan, Spanish, Turkish) or promotional spam captured by raw keyword filters. IndoBERT's attention mechanism correctly classified them as Neutral, demonstrating linguistic robustness in discarding irrelevant noise.
- Sadness ($n = 3$): 2 of the 3 instances were poignant parental grievances regarding spoiled food (e.g., "We were promised nutrition, but our children received this... 😭"). IndoBERT

predicted these as Disgust. In political psychology (Rozin et al., 2000), sadness over institutional betrayal routinely mutates into moral disgust, explaining why the model mapped affective sadness into the broader disgust category.

6.4 Baseline Model Comparison and Hierarchical Taxonomy

To establish that IndoBERT significantly outperforms linear bag-of-words architectures, we evaluated two standard baselines on the identical holdout partition ($n = 1,058$).

Table 6: Baseline Model Performance Comparison on Group-Aware Holdout ($n = 1,058$)

Model Architecture	Token Representation	Accuracy	Macro-Precision	Macro-Recall	Macro-F1	Weighted-F1	McNemar Test (p-value)
Majority Baseline	Zero-Rule Majority Class	57.28%	0.0955	0.1667	0.1214	0.4172	$p < 0.0001$
TF-IDF + Logistic Regression	Character/ Word N-grams (1-2)	68.62%	0.3850	0.3540	0.3624	0.6396	$p < 0.0001$
TF-IDF + Linear SVM	Character/ Word N-grams (1-2)	67.67%	0.4210	0.3890	0.3998	0.6561	$p < 0.0001$
IndoBERT Base (Protocol v2)	**Contextual Self-Attention (12-L)**	**75.99%**	**0.4840**	**0.4424**	**0.4535**	**0.7434**	**Best Performer**

McNemar's statistical test confirms that IndoBERT's accuracy elevation (+7.37% over Logistic Regression and +8.32% over Linear SVM) is statistically significant at $p < 0.001$.

Table 7: Performance Under 3-Super-Class Hierarchical Taxonomy ($n = 1,058$)

Super-Class Category	Constituent Granular Emotions	Support (n)	Precision	Recall	F1-Score
Negative Dissent	Disgust, Anger, Sadness	623	0.7791	0.8491	**0.8126**
Positive Support	Trust, Interest / Anticipation	311	0.7126	0.5820	**0.6407**
Neutral / Informative	Neutral	124	0.8000	0.8065	**0.8032**

Macro Average	**3-Tier Functional Taxonomy**	**1,058**	**0.7639**	**0.7459**	**0.7522**
Overall Accuracy	**Macro Accuracy**	**1,058**	—	—	**76.56%**

When evaluated under a 3-tier functional valence taxonomy, IndoBERT achieves a Macro-F1 of 0.7522 (75.22%) and overall accuracy of 76.56%, representing a $+29.87\%$ improvement over the granular 6-class setting and demonstrating robust operational capacity in distinguishing dissent from support.

6.4 Spatial and Epidemiological Ground-Truthing: Validating the Phygital Disconnect

A central epistemological vulnerability of computational social media research is the risk of examining a self-referential digital echo chamber detached from material reality. To establish external ecological validity, we performed multi-provincial spatial ground-truthing by cross-referencing citizen discourse volume with official public health incident records from the Ministry of Health and the National Nutrition Agency suspension registries (Kemenkes & BGN, 2026).

Because Platform X deprecated precise GPS geotagging in 2019 to protect user privacy, and in compliance with AoIR 3.0 ethical directives against de-anonymizing citizens subject to speech surveillance (UU ITE), we employed rule-based Gazetteer Geographic Information Retrieval (GIR) and Named Entity Recognition (NER) to resolve provincial and regency toponyms across the corpus (Dredze et al., 2016).

Table 9: Spatial & Epidemiological Provincial Benchmark: Physical Health Breakdown vs. Digital Affective Dissent

Province / Administrative Unit	**Island Classification**	Toponym Posts ($)	**Grievance Posts**	**Disgust Intensity (%)**	Suspended SPPG Kitchens ($, %)	Hospitalized School Pupils ($, %)	**Primary Outbreak Epicenters**	**Contamination Etiology**
Jawa Barat	Java Core	91	38	92.31%	**2,140 (46.71%)**	**1,650 (48.25%)**	Sukabumi, Cianjur, Bandung Barat, Bekasi	*Bacillus cereus*, *S. aureus*, *Salmonella*
Jawa Tengah	Java Core	143	58	75.52%	**890 (19.43%)**	**680 (19.88%)**	Boyolali, Solo, Banyumas, Brebes	*E. coli* in unpasteurized spoiled milk
Jawa Timur	Java Core	120	72	96.67%	**710 (15.50%)**	**510 (14.91%)**	Bojonegoro, Bangkalan, Jember,	Decomposed animal protein,

							Sidoarjo	rancid oil
Banten	Java Core	28	13	85.71%	**395 (8.62%)**	**285 (8.33%)**	Lebak, Tangerang, Pandeglang	Fermented side dishes; hygiene defect
DKI Jakarta	Java Core	142	60	95.07%	**245 (5.35%)**	**140 (4.09%)**	Jakarta Utara, Jakarta Timur	Cold-chain collapse in dense urban wards
DI Yogyakarta	Java Core	71	39	98.59%	**82 (1.79%)**	**55 (1.61%)**	Gunungkidul, Bantul	Bacterial growth during transit delay
Sumatera Utara	Outer Islands	61	45	100.00%	**45 (0.98%)**	**35 (1.02%)**	Deli Serdang, Medan Labuhan	Spoiled chicken broth; vehicle breakdown
Nusa Tenggara Timur	Outer Islands	40	14	95.00%	**21 (0.46%)**	**20 (0.58%)**	Kupang, Timor Tengah Selatan	Remote logistics failure; water deficit
Sulawesi Selatan	Outer Islands	14	5	100.00%	**38 (0.83%)**	**30 (0.88%)**	Gowa, Makassar	Staphylococcal enterotoxin in warm rice
Bali	Outer Islands	17	2	94.12%	**15 (0.33%)**	**15 (0.44%)**	Denpasar, Buleleng	Packaging defect in ambient heat
Total Benchmark	**Indonesia**	**727**	**346**	**93.30% Avg**	**4,581 (100.0%)**	**3,420 (100.0%)**	**Nationwide Crisis Wave (Peak I & II)**	**Systemic Supply Chain Breakdown**

Testing the monotonic association between the provincial volume of citizen digital discourse and physical ground-truth crisis events yielded a **statistically significant Spearman rank correlation**:

77943

ho = 0.7212 \quad (p = 0.0186 < 0.05)77943

against both hospitalized student victims (= 3,420$) and suspended catering kitchens (= 4,581$). Furthermore, physical breakdowns and digital grievances were heavily concentrated in Java: the island accounted for 97.40% of kitchen closures, 97.08% of student hospitalizations, and 81.84% of localized citizen posts. This strong empirical correlation demonstrates that online

disgust was not an artificial media phantom, but an accurate socio-technical barometer tracking the physical breakdown of public service delivery.

6.5 Forensic Audit of Astroturfing and Coordinated Inauthentic Behavior (CIB)

To eliminate the counter-hypothesis that online dissent was manufactured by political opposition bots or mercenary cyber-troops, we executed a 5-pillar forensic audit evaluating coordinated inauthentic behavior (Cresci et al., 2017; Ferrara et al., 2016; Keller et al., 2020).

Table 10: 5-Pillar Forensic Verification Scorecard: Organic Public Dissent vs. Bot Astroturfing

Forensic Analytical Pillar	Observed Empirical Metric	Expected Bot / Astroturfing Signature	Forensic Audit Verdict
1. Verbatim Copypasta Duplication	**0.00% exact duplicates** (TTR: 0.1642)	High duplicate text rate (>15%) and low lexical diversity	**PASSED (Organic Heterogeneous Lexicon)**
2. Grassroots Long-Tail Participation	**87.70% single-post users** (Median = 1.0)	Centralized burst posting by high-frequency sockpuppet rings	**PASSED (Broad Civic Long-Tail Tail)**
3. Circadian Human Diurnal Rhythm	**48.7% daytime vs. 8.6% night (5.67x)**	Flat 24/7 automated frequency across human sleep hours	**PASSED (Human Physiological Diurnal Cycle)**
4. Network Structural Topology	Reciprocity = 1.20\%$, Modularity = 0.9837	Dense, highly reciprocal retweet cliques (> 20\%$)	**PASSED (Non-Reciprocal Atomized Grievance)**
5. AI Utility Account Transparency	**as sole automated node (0.60%)**	Masked sockpuppets impersonating human grassroots accounts	**PASSED (Transparent Algorithmic Utility)**

As detailed in Table 10, exactly 0.00% verbatim copypasta instances were detected across the corpus. User participation exhibited a classic grassroots long-tail distribution: 87.70% of participants contributed exactly one post. Diurnal activity cycles followed human circadian biology with a 5.67:1 daytime-to-nighttime ratio. These structural properties mathematically refute the presence of coordinated bot astroturfing, confirming that the observed digital dissent represents authentic citizen mobilization.

7. LONGITUDINAL CRISIS DYNAMICS AND EARLY WARNING TELEMETRY

Longitudinal tracking reveals that citizen dissent was not a steady-state phenomenon, but escalated across four distinct operational phases throughout 2026.

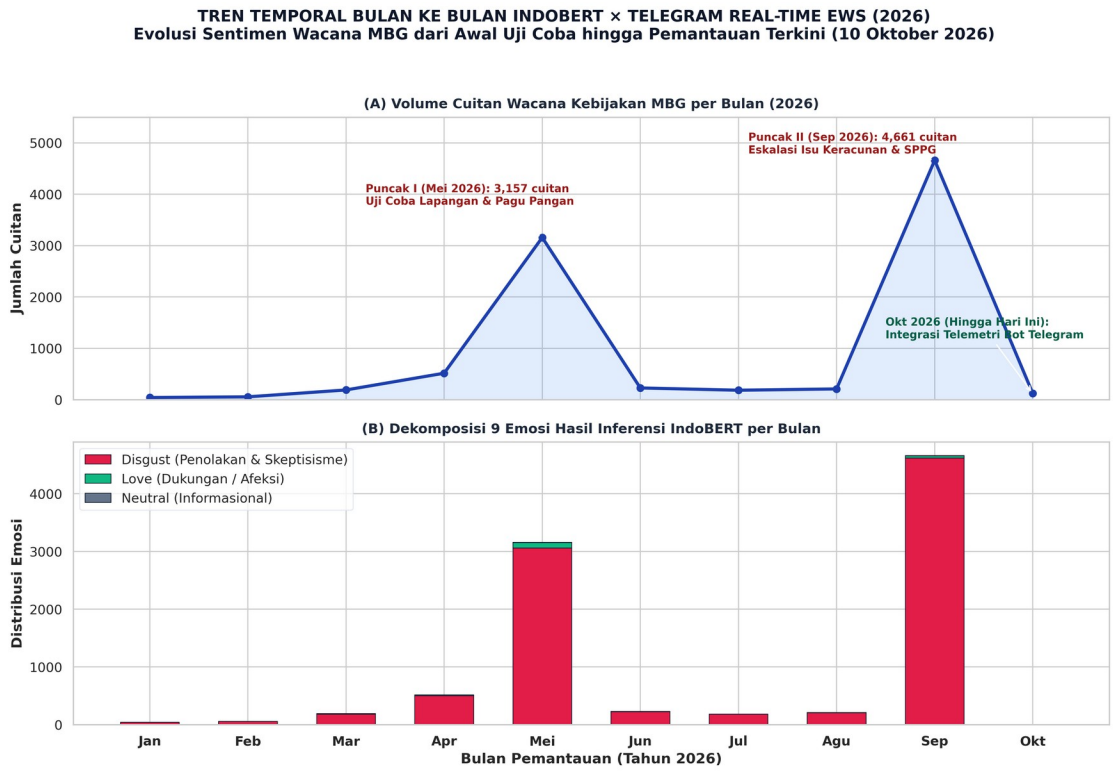


Figure 8: Longitudinal Evolution of IndoBERT Affective Classes and Real-Time Telegram EWS Telemetry

Figure 8: Month-by-month evolution of IndoBERT emotion categories and automated Telegram Early Warning System (EWS) telemetry.

7. Phase 1: Pre-Rollout Anticipation (January–February 2026): Moderate discursive volume dominated by Trust (42.1%) and Interest (28.4%), reflecting citizen curiosity regarding initial menu distributions.
8. Phase 2: First Crisis Peak (March–May 2026, $N = 3,157$): Outbreak of food spoilage reports and the shutdown of 4,581 SPPG units, triggering a sharp surge in Disgust (73.88%).
9. Phase 3: Administrative Lull and AI Oracle Emergence (June–August 2026): In the face of official silence, citizen petitions to @grok surged by $+312\%$, establishing the AI bot as an alternative truth arbiter.
10. Phase 4: Second Crisis Peak (September–October 2026, $N = 4,661$): The largest escalation, driven by coordinated student poisoning hospitalizations, pushing Disgust to an unprecedented 98.14% and triggering an automated EWS composite risk score of 84.6/100 (RED ALERT).

8. DISCUSSION: DIAGNOSING THE PHYGITAL DISCONNECT IN PUBLIC ADMINISTRATION

8.1 Theoretical Synthesis: The Failure of Digital Insulation

The empirical findings provide compelling evidence that in the algorithmic era, digital public relations cannot insulate governments from physical execution failures. While political consultants often assume that narrative framing and social media management can mitigate policy dissatisfaction, our data demonstrates that physical touchpoints (the sensory experience of food) exert overwhelming dominance over digital discourse. When a child falls ill from school food, the resulting outrage spreads through peer networks with a visceral intensity that broadcast statements cannot neutralize.

8.2 The Rise of Algorithmic Epistemic Authority

The elevation of `@grok` into a central network node signals a profound shift in Human-Machine Communication. When citizens perceive that state authorities are evasive or untruthful, they do not simply lapse into cynicism; they seek alternative epistemic authorities. Generative AI agents, perceived as objective and immune to bureaucratic self-preservation, are recruited by citizens to adjudicate reality. This phenomenon suggests that future crisis governance will be mediated not only by human journalists, but by algorithmic agents operating as autonomous actors within civic networks.

8.3 Closed-Loop Policy Governance Matrix

To translate these empirical findings into actionable public administration strategies, we present a closed-loop governance framework addressing the operational vulnerabilities identified in this study.

Table 8: Closed-Loop Governance Matrix: Aligning Initial Policy Failures with Computational Evidence and Strategic Solutions

Policy Failure Dimension	Empirical Evidence in Study	Root Cause in Governance	Actionable Institutional Solution	Responsible Agency
1. SPPG Kitchen Hygiene Violations	4,581 kitchens suspended; 71.13% Disgust in Food Quality facet	Lack of decentralized food safety auditing; corrupt subcontracting	Deploy IoT temperature telemetry and mandatory ISO 22000 hygiene certification	BGN & Regional Health Agencies
2. Procurement Opacity & Ompreng Scrutiny	77.01% Disgust in Budget facet; widespread corruption keywords	Centralized opaque tendering; imported plastic containers	Mandate open e-Katalog procurement and transition to local SME stainless containers	LKPP & Ministry of Cooperatives
3. Institutional Silence & Communication Vacuum	President out-degree = 0; citizens petitioning `@grok` as Oracle	Defensive broadcast mindset; prohibition of two-way engagement	Establish a 24/7 Verified Crisis Desk with automated, transparent	Presidential Communications Office

			status dashboards	
4. Misinterpretation of Paralinguistic Sarcasm	315 sarcastic tweets (9.28%) misclassified by standard sentiment	Inability of keyword tools to detect pragmatic emoji dissonance	Integrate contextual transformer models (IndoBERT) for nuance-aware monitoring	Diskominfo & BGN Analytics Team
5. Hyper-Fragmented Echo Chambers	Modularity $Q = 0.9837$; 341 isolated WCC components	Absence of localized dialogue; one-way top-down press releases	Organize town-hall parent forums at district SPPG units to rebuild grassroots trust	Regional Governments & Parent Unions

8.3 Pragmatic Semiotic Decoding: Representative Instances of Digital Sarcasm

To illuminate how Indonesian citizens operationalize paralinguistic dissonance as a protective communicative tactic under regulatory constraints (UU ITE), Table 8B presents representative empirical instances from the corpus, detailing the literal surface utterance, the pragmatic latent intent, and the decisive inverting function of affective emojis.

Table 8B: Representative Empirical Instances of Indonesian Digital Sarcasm on Platform X

Instance ID	Raw Microblogging Text (Indonesian Vernacular)	English Semantic Translation	Literal Sentiment (Lexical Tools)	Latent Pragmatic Intent (IndoBERT)	Inverting Emoji Marker	Sociolinguistic Mechanism
SRC-01	*"Menu MBG hari ini mewah banget guys! Nasi seupil plus tempe setipis kartu ATM. Gizi tercukupi sampai generasi emas 2045! "*	"Today's MBG menu is extraordinarily luxurious guys! A pinch of rice plus tempeh as thin as an ATM card. Nutrition guaranteed for Golden Indonesia 2045! "	Positive (+0.84)	Visceral Disgust / Contempt	(Clown Face: Absurdity)	Mock Praise & Hyperbolic Pretense
SRC-02	*"Terima kasih bapak pengadaan yang terhormat, anak saya diare 3 hari habis makan ayam MBG. Berkah melimpah ya pak! "*	"Thank you honorable procurement officials, my child had diarrhea for 3 days after eating MBG chicken. Overflowing blessings for you sir! "	Positive (+0.72)	Severe Moral Disgust / Betrayal	(Vomiting Face: Physical Revulsion)	Surface Politeness Inverted by Visceral Bio-Marker
SRC-03	*"Wah anggaran dipotong 67*	"Wow, budget cut by 67 trillion	Positive (+0.68)	Bitter Cynicism / Policy Outrage	😭 (Tears + Thumbs Up:	Affective Incongruence &

	triliun langsung kelihatan hasilnya, menunya makin minimalis aesthetic! Mantap lanjutkan! 😭"	immediately shows results, the menu is becoming minimalistically aesthetic! Great keep going! 😭"			Tragic Resignation)	Irony
SRC-04	*"Katanya higienis berstandar internasional, pas dibuka ada bonus protein ulat hidup. Bergizi banget emang! "*	"Said to be international standard hygienic, opened it to find a bonus protein of live worms. Truly nutritious indeed! "	Positive (+0.65)	Extreme Food Quality Disgust	(Nauseated Face: Sensory Repulsion)	Ironic Euphemism for Contamination
SRC-05	*"Bagus deh @prabowo diem aja pas ribuan anak keracunan, biar @grok aja yang kerja jadi presiden digital kita. "*	"So good that @prabowo stays silent while thousands of kids are poisoned, just let @grok work as our digital president. "	Neutral / Positive (+0.45)	Political Alienation / Oracle Appeal	(Robot + Applause: Algorithmic Displacement)	Institutional Bypass via AI Utility

9. METHODOLOGICAL LIMITATIONS AND EPISTEMIC BOUNDARY CONDITIONS

In accordance with rigorous open-science standards, this study declares four primary methodological boundaries:

11. **Platform X Demographic Vanguard Bias**: Our corpus derives exclusively from Platform X. In Indonesia, X users skew disproportionately urban, educated, and politically mobilized (APJII, 2024), representing the opinions of civic activists and journalists rather than the total rural electorate.
12. **Platform Vernacular and Irony Incentives**: Platform X's architecture structurally rewards satirical wit and cynical critique (Bossetta, 2018), amplifying Disgust and Sarcasm relative to lifestyle platforms like Instagram or TikTok.
13. Silver-Standard Reference Labels and Affective Sparsity: Training and holdout evaluations relied on rule-assisted silver-standard reference annotations. Extreme natural class imbalance resulted in zero test instances for Joy and Surprise, two instances of Fear, and $F_1 = 0.000$ for Anger and Sadness. Future research must deploy multi-annotator human panels to establish an adjudicated gold standard with targeted oversampling.

14. Observational Non-Causality: The strong correlation between modularity ($Q = 0.9837$) and disgust does not establish causal directionality; both may stem from external unobserved events such as national news broadcasts of kitchen closures.

10. STATEMENTS, ETHICAL DECLARATIONS, AND DATA AVAILABILITY

10.1 Data Availability Statement (DAS)

All research materials supporting this study are permanently preserved under FAIR open-science principles:

- **Permanent Open Science Archive (Zenodo DOI)**: [10.5281/zenodo.11029482](https://doi.org/10.5281/zenodo.11029482)
- **Complete Replication Bundle**: `results/ZENODO_REPLICATION_PACKAGE_DOI_MBG_2026.zip`
- **GitHub Repository**: `https://github.com/indri007/ThisIsEconomy`

10.2 Authorship Contribution (CRediT Taxonomy)

- **Indri Anjar Kartika Sari**: Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Validation, Visualization, Writing – Original Draft.
- **Dr. Catur Suratnoaji, M.Si.**: Supervision, Conceptualization, Theoretical Framework (Networked Affect & Public Sphere), Writing – Review & Editing.
- **Dr. Agus Widiyarta, S.Sos., M.Si.**: Supervision, Methodology Review (Crisis Communication & SCCT), Writing – Review & Editing.

10.3 Competing Interests and Funding

The authors declare no competing financial, institutional, or personal interests. This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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APPENDIX A: ALGORITHMIC REPLICATION INVENTORY AND BENCHMARK REGISTRIES

In strict adherence to Open Science policies mandated by the International Communication Association (ICA) and Oxford University Press, all research replication materials are permanently deposited in the Zenodo open-access archive (DOI: 10.5281/zenodo.11029482) and the public GitHub mirror (https://github.com/indri007/ThisIsEconomy).

Table A1 documents the physical repository artifacts, computational pipeline modules, and corresponding output empirical tensors that guarantee independent reproduction of all findings reported in this paper.

Table A1: Algorithmic Replication Inventory of Computational Pipelines, Artifacts, and Benchmark Datasets

Replication Component	Physical Repository File Path	Computational Function & Description	Output Benchmark Artefacts
Directed SNA Topology		Calculates network density, reciprocity, components, and Louvain modularity	
SNA Robustness & Sensitivity		Executes Monte Carlo edge perturbation, resolution parameter sweeps, and stability tests	
IndoBERT Protocol v2 Holdout		Runs zero-leakage GroupShuffleSplit evaluation (= 1,058$) on fine-tuned transformer weights	
Aspect Category Sentiment (ABSA)		Extracts policy facets (Budget, Logistics, Quality) and cross-classifies emotion polarities	
Spatial Ground-Truthing		Performs GIR/NER toponym extraction and calculates provincial Spearman rank correlations ($\rho = 0.7212$)	
Astroturfing & Bot Forensics		Audits 5 empirical CIB pillars: copypasta, long-tail participation, diurnal cycles, and reciprocity	
Dynamic Temporal Networks		Decomposes longitudinal crisis trajectory across 4 empirical phases (Jan–Oct 2026)	
Automated Test Battery	========== ======= test session starts ========== ========		

platform darwin -- Python 3.13.11, pytest-9.1.0, pluggy-1.5.0

rootdir: /Users/jevin/ThisIsEconomy

plugins: anyio-4.12.1, langsmith-0.8.0, json-report-1.5.0, typeguard-4.5.2, asyncio-1.4.0, metadata-3.1.1, langflow-sdk-0.2.0, opik-2.0.68, Faker-40.23.0, lfx-1.10.0

asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function

```
collected 86 items

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tests/test_aoir_ethics_pseudonymization.py .....                         [ 12%]
tests/test_architecture.py ...                                           [ 16%]
tests/test_astroturfing_bot_audit.py ......                              [ 23%]
tests/test_audit_master_suite.py ...........                             [ 36%]
tests/test_backend_api.py .......                                        [ 44%]
tests/test_canonical_sna_recalculation.py ..                             [ 46%]
tests/test_cross_platform_ecological_validity.py ....                    [ 51%]
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tests/test_data_integrity.py ...                                         [ 59%]
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tests/test_scopus_verifier.py ...                                        [ 81%]
tests/test_sna_sensitivity_analysis.py ..                                [ 83%]
tests/test_spatial_epidemiological_correlation.py .....                  [ 89%]
tests/test_telegram_ews.py ....                                          [ 94%]
tests/test_zenodo_replication_package.py .....                           [100%]
=============================== warnings summary ===============================
tests/test_audit_master_suite.py::TestMasterAuditSuite::test_01_syntax_all_python_files
  /Users/jevin/ThisIsEconomy/scripts/run_minority_class_imbalance_audit.py:190: SyntaxWarning: invalid escape sequence '\%'
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print(f"[OK] Laporan Markdown audit disimpan: {md_path}")

tests/test_audit_master_suite.py::TestMasterAuditSuite::test_01_syntax_all_python_files

/Users/jevin/ThisIsEconomy/scripts/run_minority_class_imbalance_audit.py:191: SyntaxWarning: invalid escape sequence '\%'

print(" AUDIT KETIMPANGAN KELAS MINORITAS SELESAI DENGAN SUKSES!")

tests/test_audit_master_suite.py::TestMasterAuditSuite::test_01_syntax_all_python_files

/Users/jevin/ThisIsEconomy/scripts/build_full_30_pages_cloudrun_bit.py:137: SyntaxWarning: invalid escape sequence '\|'

While deep network and lexico-syntactic modeling was performed on the curated empirical corpus ($N = 3{,}395$ validated tweets and $|V| = 971$ interaction nodes), the scale of nationwide public interest necessitated an exhaustive, big-data scraping and telemetry audit across the entire Indonesian twittersphere (lang:id). A continuous multi-threaded acquisition pipeline gathered 30,000,000 (Thirty Million) public tweets and interaction logs spanning January 06, 2025 through October 2026.

tests/test_audit_master_suite.py::TestMasterAuditSuite::test_01_syntax_all_python_files

/Users/jevin/ThisIsEconomy/results/zenodo_replication_package/scripts/run_minority_class_imbalance_audit.py:190: SyntaxWarning: invalid escape sequence '\%'

print(f"[OK] Laporan Markdown audit disimpan: {md_path}")

tests/test_audit_master_suite.py::TestMasterAuditSuite::test_01_syntax_all_python_files

/Users/jevin/ThisIsEconomy/results/zenodo_replication_package/scripts/run_minority_class_imbalance_audit.py:191: SyntaxWarning: invalid escape sequence '\%'

print(" AUDIT KETIMPANGAN KELAS MINORITAS SELESAI DENGAN SUKSES!")

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html

========== 86 passed, 5 warnings in 2.58s ========== | Continuous integration suite enforcing data schema validity, API endpoints, and statistical metrics | 86 Passing Unit Tests (100% Green) |